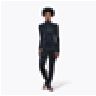

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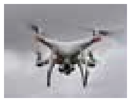


UK vs drones
The UK government will introduce draft legislation aimed at preventing unsafe or criminal use of drones. **http://dgit.in/UKvDrone**

Credit: NASA/JPL-Caltech

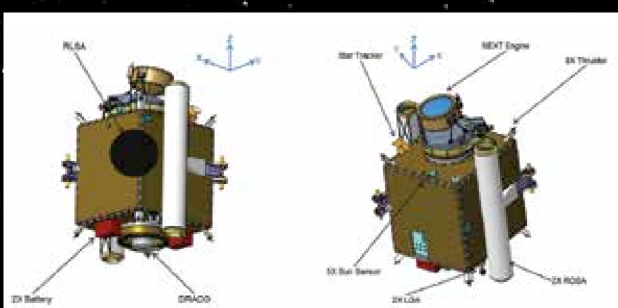
The NEOWISE is an asteroid hunter that scours the universe for potentially dangerous objects hurling towards Earth

plans to defending ourselves from an incoming asteroid has always been an agenda since we discovered that any object from outer space can simply drop in and annihilate everything. The only two ways to go forward from protecting ourselves is to destroy the asteroid or deflect its path.

The first and obvious measure would be to destroy the asteroid before it comes near. Since asteroids are huge, you will need a really powerful bomb and nothing is stronger than a nuclear explosion. Point the nuke towards the asteroid and blow it up, easy. However, it isn't as simple as it sounds. This might not work because we don't know whether the asteroid will blow up to smithereens or just break ups into pieces. The debris will still continue a crash course to Earth and with more objects, the damage might be scaled up. So, it's a big no to blowing up incoming space rocks.

Following Newton's first law of motion, if an asteroid travelling at a high speed collides with another object, the impact will change its path. So, instead of completely blowing up the asteroid, we could send a probe (known as kinetic impactors) right into the asteroid and change its trajectory enough to avoid collision. Of course, there's a lot of calculation required to accomplish this like figuring out the speed, trajectory, and size of the asteroid, finding the optimal location for impact, etc. We wouldn't want to mess this up otherwise the deflection might just change the course of the asteroid to a new location on Earth. NASA's Double Asteroid Redirection Test (DART) is an ongoing mission to demonstrate the kinetic impact technique for the first time.

Another way to deflect the path of hazardous asteroids is by using gravitational tractors. These don't exist at the moment obviously but are part of a number of theoretical methods being considered to effectively tackle an impending asteroid thread. In this method a probe blasts off towards the asteroid and starts orbiting around it. The probe will exert a minor magnitude of gravity on the rock and after a long period of time, the course of the asteroid would have changed. If that seems too slow, a rocket can be sent to directly attach itself to the asteroid and slowly pull it out of its regular orbit. One shortcoming to these methods is we'd have to be aware about the potential trajectory way before so that we can have enough time to send the probe.

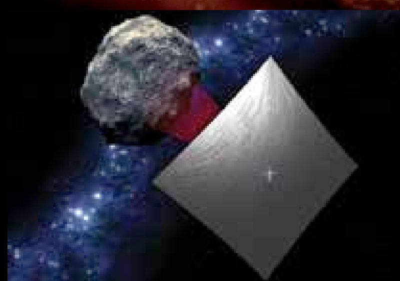


Credit: NASA

DART Mission to demonstrate kinetic impactors

Another method being considered is the use of solar sails. These take advantage of the phenomenon called solar or radiation pressure, where the photons from the Sun pushes any object in space. It's a form of spacecraft propulsion where the light will fall on large mirrors, eventually applying enough pressure to accelerate the object. This phenomenon can be applied on an incoming asteroid by tethering a solar sail system around the asteroid and ultimately changing its course. This method also requires an advance intimation about the incoming asteroid.

Many asteroids have frozen materials present on its surface and sometimes they are completely made of a frozen substance. When they enter the Earth's atmosphere, they evaporate and long tails are formed. Similarly, by using powerful lasers or maybe focusing sunlight on the asteroid, the frozen materials can be melted or evaporated. Eventually, until it reaches the Earth, it would either completely deteriorate or reduced in size which won't be a threat anymore.



Credit: NASA

Solar sails will make it easier to change the asteroid's trajectory

In conclusion, if an asteroid was to impact Earth in the next few years, we are almost defenceless. With effort being made to identify potentially dangerous objects approaching for impact, we might be able to discover threats way in advance. But unless we identify all the hazardous objects lurking in space, it's difficult to prepare against them. Meanwhile, a couple of missions and projects are underway to try out different techniques that could divert the danger of humankind going extinct. d